rootstratum.com

AWS Infrastructure Architecture

Cloud Provider: AWS | Region: ap-south-1 (Mumbai) | Account: 865780387311
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Infrastructure	AWS EC2 + RDS + VPC + IAM
Compute	t4g.small — Graviton2 arm64 (Reserved, 1yr No Upfront)
Database	RDS PostgreSQL 16 — db.t4g.micro (Free Tier 12 mo)
Networking	VPC 10.0.0.0/16 — 3 Public + 3 Private Subnets across 3 AZs
Application	Next.js (port 3000) + FastAPI (port 8011) behind Nginx
SSL / TLS	Let's Encrypt via Certbot — auto-renewed on EC2
DNS	Cloudflare — A record → Elastic IP
CI / CD	GitHub Actions + OIDC (keyless) → SSM SendCommand
IaC	Terraform with S3 remote state + DynamoDB lock

Architecture Diagram

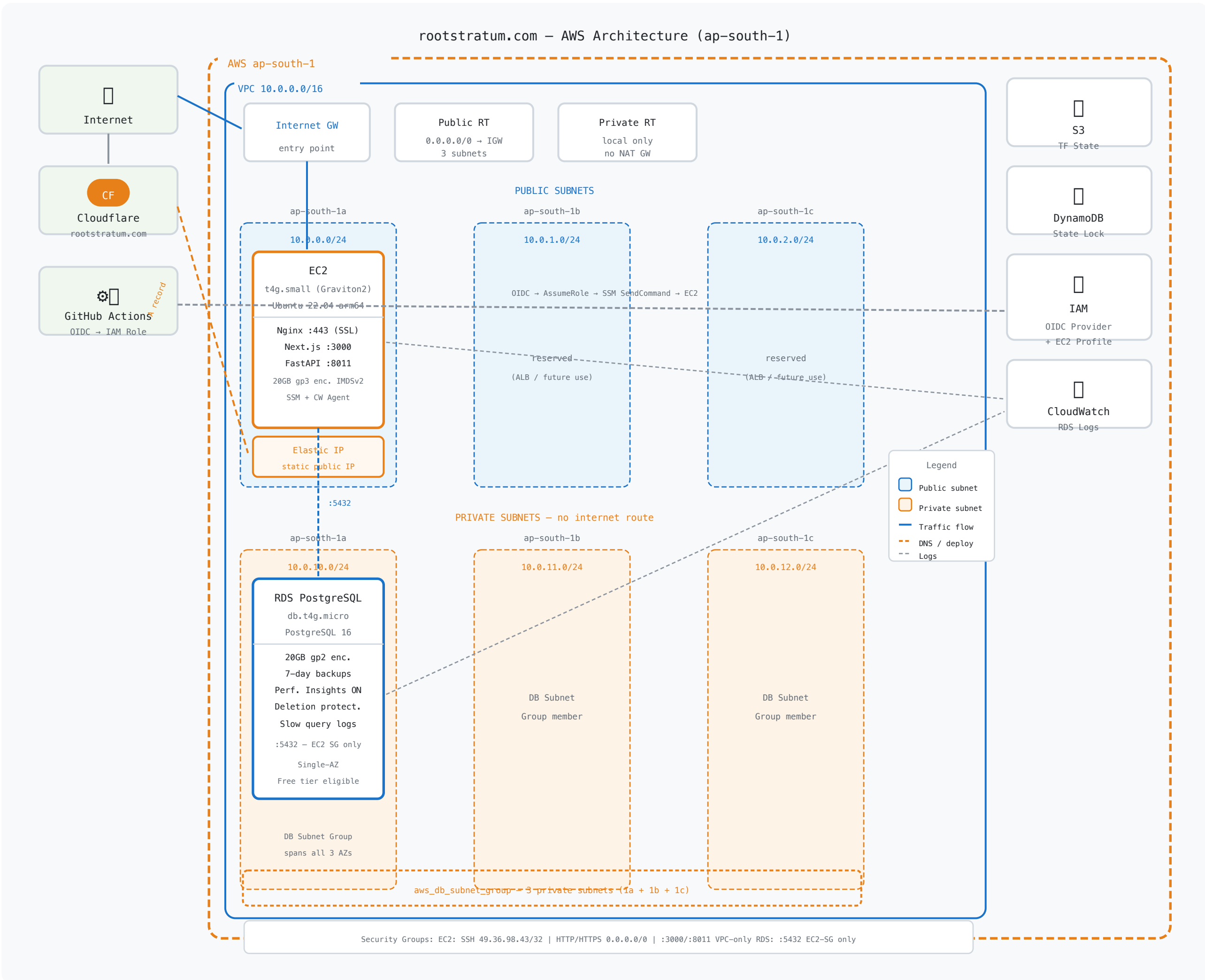

Figure 1 — rootstratum.com full AWS architecture across 3 Availability Zones in ap-south-1.

Architecture Explained

1. Networking — VPC & Subnets

All resources live inside a dedicated Virtual Private Cloud (VPC) with CIDR **10.0.0.0/16**, spanning all three Availability Zones in ap-south-1 (1a, 1b, 1c). This layout provides the foundation for high availability without any single point of failure at the network layer.

Public Subnets (10.0.0.0/24, 10.0.1.0/24, 10.0.2.0/24)

Three public subnets, one per AZ, are attached to a single Internet Gateway via a shared Public Route Table. The EC2 app server sits in the 1a public subnet with an Elastic IP providing a static entry point. The 1b and 1c public subnets are reserved for future use — an Application Load Balancer can be added across all three with zero networking changes.

Private Subnets (10.0.10.0/24, 10.0.11.0/24, 10.0.12.0/24)

Three private subnets, one per AZ, have only a local route — no NAT Gateway. This keeps costs minimal (NAT Gateway costs ~$32/month plus data transfer) while keeping the database completely unreachable from the internet. The RDS DB Subnet Group spans all three private subnets, so a future Multi-AZ promotion requires no networking changes.

2. Compute — EC2 t4g.small

The application runs on a single **t4g.small** EC2 instance (2 vCPU, 2 GB RAM) powered by AWS Graviton2 (arm64). This instance class delivers up to 40% better price-performance than equivalent x86 types. It is purchased as a **1-year Standard Reserved Instance (No Upfront)** at ~$7.50/month versus ~$13.50/month on-demand — a 44% saving.

OS	Ubuntu 22.04 LTS arm64
Root volume	20 GB gp3 — encrypted at rest
IMDSv2	Required — blocks SSRF-based credential theft
Elastic IP	Static public IP assigned — Cloudflare A record points here
IAM Profile	AmazonSSMManagedInstanceCore + CloudWatchAgentServerPolicy
Processes	Nginx (SSL termination), Next.js :3000, FastAPI :8011

Security Group Rules

Port / Protocol	Source	Purpose
22 / TCP	49.36.98.43/32	SSH — admin IP only
80 / TCP	0.0.0.0/0	HTTP (Nginx redirects to HTTPS)
443 / TCP	0.0.0.0/0	HTTPS — public traffic
3000 / TCP	VPC CIDR (10.0.0.0/16)	Next.js — internal only, proxied by Nginx
8011 / TCP	VPC CIDR (10.0.0.0/16)	FastAPI — internal only, proxied by Nginx

3. Database — RDS PostgreSQL 16

The database is fully decoupled from the application server, running as a managed **Amazon RDS** instance. This separation means database operations (backups, patching, failover) do not affect EC2 availability, and the EC2 can be replaced without touching the data layer.

Engine	PostgreSQL 16
Instance class	db.t4g.micro — free tier eligible for 12 months
Storage	20 GB gp2 — encrypted at rest (AES-256)
Multi-AZ	Disabled (single-AZ for cost) — promote to Multi-AZ any time
Public access	Disabled — reachable only from the EC2 security group on :5432
Backups	7-day retention, daily window 02:00–03:00 UTC
Performance Insights	Enabled — 7-day free retention, query-level visibility
Slow query logging	Queries >1s logged to CloudWatch (log_min_duration_statement)
Connection audit	log_connections and log_disconnections enabled
Deletion protection	Enabled — Terraform cannot destroy the DB without code change
Auto minor upgrades	Enabled — security patches applied automatically
Parameter group	Custom pg16 group: pg_stat_statements, slow query, audit logs

4. IAM & Security

All IAM access follows the principle of least privilege. The root account is not used for day-to-day operations — an IAM admin user (**rootstratum-admin**) was created via the bootstrap phase. Two separate IAM roles handle runtime access.

EC2 Instance Role

- Attached via an IAM Instance Profile to the EC2 — credentials are never stored on disk.
- **AmazonSSMManagedInstanceCore** — registers the instance with SSM, enabling GitHub Actions to run deploy commands without SSH.
- **CloudWatchAgentServerPolicy** — allows the CloudWatch agent to ship logs and metrics.

GitHub Actions OIDC Role

- GitHub Actions authenticates via **OpenID Connect (OIDC)** — no long-lived access keys are stored in GitHub Secrets.
- Trust policy is scoped to **refs/heads/main only** — pull requests cannot assume this role.
- SSM SendCommand permission is further restricted by resource tags (*Project=rootstratum, Environment=prod*) — the role cannot accidentally target other AWS resources.
- GitHub Actions deploys by calling **SSM SendCommand** → EC2 runs the deploy script. Port 22 never needs to be opened to GitHub's IP ranges.

5. CI/CD Pipeline & DNS

Deployments are fully automated via GitHub Actions triggered on every push to the **main** branch. The pipeline assumes the IAM role via OIDC, then issues an SSM SendCommand to run the deploy script on the EC2 instance. No SSH keys are stored in GitHub.

DNS is managed on **Cloudflare**. A single A record points **rootstratum.com** to the Elastic IP. Because the Elastic IP is static, the DNS record never needs updating even after EC2 stop/start cycles. SSL is terminated on the EC2 by Nginx using a **Let's Encrypt** certificate managed by Certbot.

6. Terraform State Management

Infrastructure state is stored remotely in **S3** (*rootstratum-tfstate-865780387311*) with versioning and AES-256 encryption enabled. A **DynamoDB** table provides state locking to prevent concurrent applies from corrupting state. Both the S3 bucket and DynamoDB table were created by the **bootstrap** module, which runs once with local state before the main infrastructure is provisioned.

7. Estimated Monthly Cost

Resource	Spec	Est. Cost/mo
EC2 t4g.small	Reserved 1yr No Upfront	~$7.50
Elastic IP	Attached to running instance	~$0.00
EBS gp3 20GB	Root volume	~$1.60
RDS db.t4g.micro	Free tier (12 months)	$0 → ~$12
RDS storage 20GB	gp2 — free tier	$0 → ~$2.30
S3 state bucket	Minimal state files	~$0.01
DynamoDB lock table	Pay-per-request	~$0.00
Data transfer out	First 100GB/mo free	~$0.00
CloudWatch logs	RDS PostgreSQL logs	~$0.50
		~$10 / ~$24

Costs are estimates in USD for ap-south-1. RDS free tier applies for 12 months on new accounts. EC2 Reserved Instance pricing as of 2025.